Introduction to Compressed Sensing. The Restricted Isometry Property

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Abstract

In this chapter, we introduce the *Null Space Property* and the *Restricted Isometry Property* for measurement matrices. These conditions imply the Exact Reconstruction condition $RE(s)$ of a matrix. We then prove that Gaussian random matrices satisfy these properties with high probability for the optimal number of measurements of order $s\log(eN/s)$. This shows that if one takes $m \sim s\log(eN/s)$ Gaussian measurements of an s-sparse vector in $\mathbb{R}^N$, then one can reconstruct it exactly by the Basis Pursuit procedure.

1 Introduction

The Basis Pursuit procedure was introduced by convex relaxation in the previous lecture. We recall its definition.

Definition 1.1 ([2, 1]). *Given a matrix $A \in \mathbb{R}^{m\times N}$ and $y \in \mathbb{R}^m$, the **Basis Pursuit** (BP) procedure returns*

$$\hat{x} \in \operatorname*{argmin}_{t\in\mathbb{R}^N : At=y} \|t\|_1 \tag{BP}$$

when there is a solution to the equation $At = y$, and $\emptyset$ otherwise.

We have seen that one can rewrite (BP) as a linear programming problem, and therefore implement this procedure efficiently even in high dimension N.

From a theoretical viewpoint, we have shown that if a measurement matrix $A \in \mathbb{R}^{m\times N}$ allows the reconstruction of any signal in Σ_s by Basis Pursuit, then necessarily $m \gtrsim s\log(eN/s)$, i.e., one must have at least $s\log(eN/s)$ linear measurements in order to solve the Compressed Sensing problem by BP on Σ_s.

In this chapter, we introduce conditions on the matrix A that guarantee exact reconstruction by BP of every vector in Σ_s. The property we seek is recalled here.

Definition 1.2. *Let $A : \mathbb{R}^N \to \mathbb{R}^m$ be a matrix such that $m \leq N$, and let s be an integer smaller than N. We say that A satisfies the property of **exact reconstruction of order s** when, for every s-sparse vector x,*

$$\operatorname*{argmin}_{t\in\mathbb{R}^N} \big(\|t\|_1 : At = Ax \big) = \{x\}.$$

We then say that A satisfies $RE(s)$.

We already know that if A satisfies $RE(s)$, then A necessarily has at least $s\log(eN/s)$ rows (always up to an absolute constant). The goal of this chapter is to show that one can construct matrices satisfying $RE(s)$ and having a *minimal* number of rows, i.e., of the order of $s\log(eN/s)$. We will see that random matrices satisfy these properties.

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2 Null Space Property and Restricted Isometry Property

In the previous chapter, we saw that the kernel of A plays an important role for the ℓ_0-minimization procedure: this procedure solves the CS problem on Σ_s if and only if $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$.

The only element of $\mathrm{Ker}(A)$ that is allowed to be sparse is therefore 0. This condition is also necessary simply in order to have $Ax \neq Ay$ for every $x \neq y \in \Sigma_s$, i.e., in order for the CS problem to have a chance of being solved on Σ_s.

In this section, we show that $\mathrm{Ker}(A)$ also plays an essential role for the property $RE(s)$. We want $\mathrm{Ker}(A)$ to lie, in some sense, far from Σ_{2s}, and not merely to have zero intersection with Σ_{2s}. This property is known in CS under the name *Null Space Property*. We recall that if $v \in \mathbb{R}^N$ and $J \subset \{1, \dots, N\}$, then v_J is the vector in $\mathbb{R}^N$ that coincides with v on J and has zero coordinates outside J.

Definition 2.1. *Let $A \in \mathbb{R}^{m \times N}$ and $1 \leq s \leq N$. We say that A satisfies the **Null Space Property of order** s when, for every $J \subset \{1, \dots, N\}$ such that $|J| = s$ and every $v \in \mathrm{Ker}(A) \backslash \{0\}$, one has:*

$$\|v_J\|_1 < \|v_{J^c}\|_1 \tag{2.1}$$

We then say that A satisfies $NSP(s)$.

Intuitively, condition (2.1) says that if v is a nonzero element of $\mathrm{Ker}(A)$, then there cannot exist s coordinates of v that contain more than half of its "mass" in the ℓ_1 sense. One can rewrite (2.1) as $2\|v_J\| < \|v\|_1$, and this for every $J \subset \{1, \dots, N\}$ of cardinality s. In particular, one may choose the s largest coefficients of v in absolute value, and the condition $NSP(s)$ is thus equivalent to: for every $v \in \mathrm{Ker}(A) \backslash \{0\}$

$$v_1^* + \cdots + v_s^* < \frac{\|v\|_1}{2}$$

where $v_1^* \geq v_2^* \geq \cdots \geq v_N^* \geq 0$ is the nonincreasing rearrangement of $(|v_j|)_{j=1}^N$. In some sense, the nonzero vectors of $\mathrm{Ker}(A)$ must be the opposite of sparse vectors; one says that they are "well spread".

In particular, nonzero $2s$-sparse vectors cannot lie in $\mathrm{Ker}(A)$, since if $v \in \Sigma_{2s}$ then $v_1^* + \cdots + v_s^* \geq v_{s+1}^* + \cdots + v_{2s}^* = \|v\|_1 - (v_1^* + \cdots + v_s^*)$. In particular, the Null Space Property implies the necessary and sufficient condition for exact reconstruction by ℓ_0.

Proposition 2.2. *Let $A \in \mathbb{R}^{m \times N}$. If A satisfies $NSP(s)$, then* $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$.

The condition $NSP(s)$ is stronger than the condition $\mathrm{Ker}(A) \cap \Sigma_{2s} = \{0\}$ of the previous chapter. Vectors that are "approximately" s-sparse cannot lie in $\mathrm{Ker}(A)$ either under $NSP(s)$; for example, if A satisfies $NSP(s)$, then

$$v = \Big(\underbrace{1, \dots, 1}_{s \text{ times}}, \underbrace{\frac{s}{N-s}, \dots, \frac{s}{N-s}}_{N-s \text{ times}} \Big)^\top \notin \mathrm{Ker}(A) \tag{2.2}$$

since $\|v_J\|_1 = \|v_{J^c}\|_1$ for $J = \{1, \dots, s\}$. In other words, v is not sufficiently "spread out": it concentrates more than half of its mass on only s coordinates, and therefore cannot lie in the kernel of A when A satisfies $NSP(s)$.

We now show that the Null Space Property $NSP(s)$ is a necessary and sufficient condition for A to satisfy $RE(s)$.

Theorem 2.3. *Let $A \in \mathbb{R}^{m\times N}$. The following two assertions are equivalent:*

1. *A satisfies $RE(s)$*
2. *A satisfies $NSP(s)$.*

Proof. If A satisfies $RE(s)$, then for every $x \in \Sigma_s$, x is the unique element of $x + \mathrm{Ker}(A)$ of minimal ℓ_1-norm, and therefore for every $v \in \mathrm{Ker}(A)\backslash\{0\}$, $\|x\|_1 < \|x+v\|_1$. Thus, for every $v \in \mathrm{Ker}(A)\backslash\{0\}$ and every $J \subset \{1,\dots,N\}$ such that $|J| = s$, taking $x = -v_J \in \Sigma_s$, one has

$$\|v_J\|_1 = \|x\|_1 < \|x+v\|_1 = \|v_{J^c}\|_1 .$$

Since this holds for every $v \in \mathrm{Ker}(A)\backslash\{0\}$ and every $J \subset \{1,\dots,N\}$ such that $|J| = s$, we deduce that A satisfies $NSP(s)$.

Conversely, assume that A satisfies $NSP(s)$. Let $x \in \Sigma_s$. Take $J \subset \{1,\dots,N\}$ such that $|J| = s$ and $\mathrm{supp}(x) \subset J$. Let $v \in \mathrm{Ker}(A)\backslash\{0\}$. One has

$$\|x+v\|_1 = \|v_{J^c}\| + \|v_J + x\|_1 \geq \|x\|_1 + \|v_{J^c}\|_1 - \|v_J\|_1 > \|x\|_1 .$$

Therefore A satisfies $RE(s)$. ∎

The condition $NSP(s)$ is therefore a necessary and sufficient condition for a matrix A to satisfy the property $RE(s)$. From a theoretical viewpoint, it now suffices to construct measurement matrices A satisfying $NSP(s)$ and having a minimal number of rows (i.e., of order $m \sim s\log(eN/s)$) in order to solve the CS problem on Σ_s via the Basis Pursuit procedure.

The condition $NSP(s)$ is not easy to check directly, however. Concretely, we do not know how to construct matrices A for which one verifies $NSP(s)$ directly. We will therefore resort to two other conditions on A that imply $NSP(s)$ and that we will be able to prove. These two conditions are sufficient—unlike NSP, which is also necessary—for having $RE(s)$. For the cases of interest in this course, these are the conditions we will seek to prove.

The first of these two conditions is related to results in the asymptotic theory of Banach spaces. It concerns what are called the *Euclidean sections of* B_1^N (we refer the interested reader to the chapter devoted to these questions for further details). For the moment, the following condition says that the kernel of A intersects the unit ball B_1^N of the ℓ_1^N-norm in a very particular way.

To fix the notation, we recall that for every $p \geq 1$, the unit ball of the ℓ_p^N-norm is denoted $B_p^N = \{x \in \mathbb{R}^N : \|x\|_p \leq 1\}$, and since $\|x\|_2 \leq \|x\|_1 \leq \sqrt{N}\,\|x\|_2$ for every $x \in \mathbb{R}^N$ (since it is clear that $\sum_j |x_j||x_j| \leq \left(\sum_j |x_j|\right)^2$ and, by Jensen, $\sqrt{(1/N)\sum_j x_j^2} \leq (1/N)\sum_j |x_j|$), one has

$$\frac{1}{\sqrt{N}} B_2^N \subset B_1^N \subset B_2^N. \tag{2.3}$$

Both bounds are attained in (2.3), for instance by $(\pm 1/N,\dots,\pm 1/N)^\top$ and $(\pm 1, 0,\dots,0)^\top$ respectively. In particular,

$$\mathrm{diam}(B_1^N, \ell_2^N) := \sup_{x\in B_1^N} \|x\|_2 = 1$$

and this bound is attained, for instance, at $(1,0,\dots,0)^\top$ (here one speaks of a diameter, whereas it is rather a radius; up to a factor 2, however, the quantities are of the same order of magnitude).

The ℓ_2^N-diameter of B_1^N is attained on the canonical axes of $\mathbb{R}^N$, which are in some sense the sparsest vectors of $\mathbb{R}^N$, since they have only one nonzero coordinate.

Under $NSP(s)$, the kernel $\mathrm{Ker}(A)$ does not contain, in particular, any sparse vector of Σ_{2s}—apart from 0—so $\mathrm{Ker}(A)$ lies far from the canonical axes and from all the $2s$-dimensional spaces spanned by these vectors. Intuitively, $\mathrm{Ker}(A)$ is positioned along *diagonal directions* of $\mathbb{R}^N$. For these vectors, which are said to be *well-spread*, the ratio of their ℓ_2-norm to their ℓ_1-norm is much smaller than 1. One may therefore expect $NSP(s)$ to hold if the ℓ_2-norm of the vectors of B_1^N that lie in the kernel $\mathrm{Ker}(A)$ is much smaller than 1. This is what is represented in Figure 2 and formalized in the following definition.

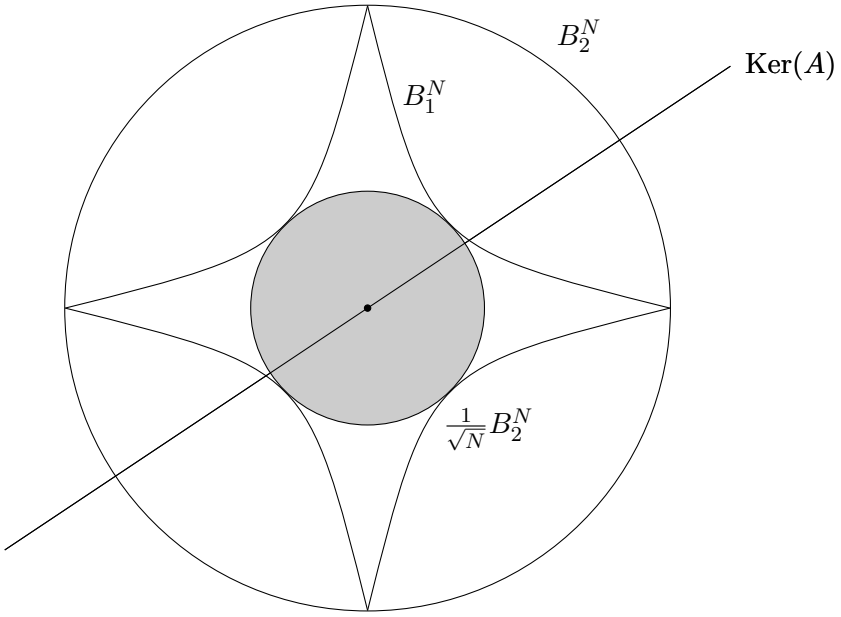


Figure 1: One has $\mathrm{diam}(B_1^N, \ell_2^N) = 1$. If $\mathrm{diam}(B_1^N \cap \mathrm{Ker}(A), \ell_2^N) < 1/(2\sqrt{s})$, then $NSP(s)$ is satisfied by A.

Definition 2.4. *Let $A \in \mathbb{R}^{m\times N}$ and $1 \leq s \leq N$. We say that A satisfies the **Gelfand property of order** s when*

$$\mathrm{diam}(B_1^N \cap \mathrm{Ker}(A), \ell_2^N) < \frac{1}{2\sqrt{s}}.$$

We then say that A satisfies $Gelfand(s)$.

The dimension of the kernel $\mathrm{Ker}(A)$ is at least $N-m$. One usually has $m << N$, so $\mathrm{Ker}(A)$ is a linear space of very large dimension, almost equal to N. We have seen that $\mathrm{diam}(B_1^N, \ell_2^N) = 1$; here the condition $Gelfand(s)$ "requires" that $\mathrm{diam}(B_1^N \cap \mathrm{Ker}(A), \ell_2^N) \leq 1/(2\sqrt{s})$, i.e., that the restriction of B_1^N to $\mathrm{Ker}(A)$ have a diameter much smaller than 1 even though $\mathrm{Ker}(A)$ has very large dimension. It is therefore a very restrictive condition on $\mathrm{Ker}(A)$. We will nevertheless prove below that random matrices can achieve it (with high probability). For the moment, we show that it is sufficient to prove the Null Space Property.

Theorem 2.5. *Let $A \in \mathbb{R}^{m\times N}$ and $1 \leq s \leq N$. If A satisfies $Gelfand(s)$, then A satisfies $NSP(s)$.*

Proof. Assume that A satisfies $Gelfand(s)$. Then for every $v \in \mathrm{Ker}(A)\backslash\{0\}$, one has $\|v\|_2 < \|v\|_1/(2\sqrt{s})$, since $v/\|v\|_1 \in B_1^d \cap \mathrm{Ker}(A)$ and therefore $\|v/\|v\|_1\|_2 < 1/(2\sqrt{s})$.

Let $v \in \mathrm{Ker}(A)\backslash\{0\}$ and $J \subset \{1,\dots,N\}$ such that $|J| = s$. One has

$$\|v_J\|_1 \leq \sqrt{s}\,\|v_J\|_2 \leq \sqrt{s}\,\|v\|_2 < \sqrt{s}\frac{\|v\|_1}{2\sqrt{s}} = \frac{\|v\|_1}{2}.$$

■

The Gelfand property remains difficult to prove directly, however, because it is a property of the kernel of A. It is in general easier to study the action of A on $\mathbb{R}^N$, or on some of its subsets, than on its kernel. We then introduce a third and last property, concerning the action of A on s-sparse vectors. The idea is that if A acts almost as an isometry on Σ_s, then $\mathrm{Ker}(A)$ cannot pass through the s-sparse vectors (other than 0) and must even stay sufficiently far from them. This is precisely the type of property one wishes to see satisfied by the measurement matrix A. It is this last property that has been the most successful in CS for proving $RE(s)$, and we state it now.

Definition 2.6. *Let $A \in \mathbb{R}^{m\times N}$ and $1 \leq s \leq N$. We say that A satisfies the **Restricted Isometry Property of order** s when, for every $x \in \Sigma_s$,*

$$\frac{1}{2}\|x\|_2 \leq \frac{\|Ax\|_2}{\sqrt{m}} \leq \frac{3}{2}\|x\|_2$$

We then say that A satisfies $RIP(s)$.

The RIP condition is the most popular one in CS. If A satisfies $RIP(s)$, then it acts as an isometry (the correct term would be an *isomorphism*, because of the constants 1/2 and 3/2) on the set of vectors Σ_s.

We recall that A is a compression matrix, and therefore has a kernel $\mathrm{Ker}(A)$ of very large dimension in $\mathbb{R}^N$. In particular, A cannot be an isometry/isomorphism on the whole space $\mathbb{R}^N$. What $RIP(s)$ requires is that A act as an isometry restricted to a subset of $\mathbb{R}^N$, here Σ_s.

Remark 2.7. *The conditions $RE(s)$, $NSP(s)$ and $Gelfand(s)$ involve A only through its kernel. In particular, these properties are unchanged under any rescaling of A, since $\mathrm{Ker}(\lambda A) = \mathrm{Ker}(A)$ for every $\lambda \neq 0$. In the case of the condition $RIP(s)$, the matrix A appears directly, and the normalization therefore plays a role in its definition.*

In Definition 2.6, we chose to regard the matrix A as the matrix of measurement vectors, i.e., the matrix whose rows are the measurement vectors $X_1,\dots,X_m$:

$$A = \begin{pmatrix} X_1^\top \\ \vdots \\ X_m^\top \end{pmatrix}$$

For this type of matrix (without rescaling), one states the condition $RIP(s)$ in which the term $\|Ax\|^2/\sqrt{m}$ is compared with $\|x\|_2$.

There is also another classical normalization in CS, which consists in scaling the matrix A by $1/\sqrt{m}$:

$$\Gamma = \frac{1}{\sqrt{m}}A = \frac{1}{\sqrt{m}}\begin{pmatrix} X_1^\top \\ \vdots \\ X_m^\top \end{pmatrix}.$$

For this rescaling, $RIP(s)$ consists in comparing $\|\Gamma x\|_2$ (without the factor $1/\sqrt{m}$) with $\|x\|_2$: for every $x \in \Sigma_s$, $(1/2)\|x\|_2 \leq \|\Gamma x\|_2 \leq (3/2)\|x\|_2$.

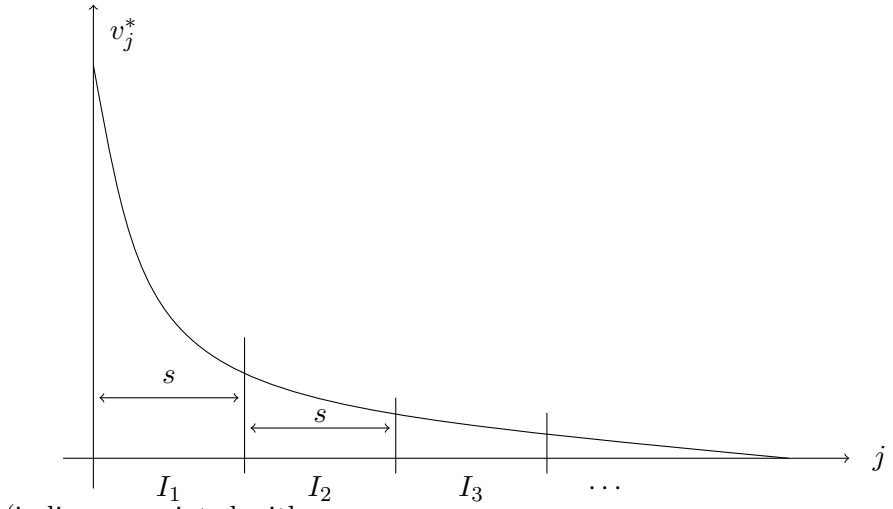
A key result in CS is that, in order to construct a measurement matrix satisfying the condition $RE(s)$, it suffices to check that A satisfies $RIP(s)$. Indeed, we show in the following result that $RIP(s)$ implies $Gelfand(s/65)$ (which implies $NSP(s/65)$, which is equivalent to $RE(s/65)$).

Theorem 2.8. *Let $A \in \mathbb{R}^{m\times N}$ and $1 \leq s \leq N$. If A satisfies $RIP(s)$, then A satisfies $Gelfand(s/65)$.*

Remark 2.9. *At no point will we seek to optimize the numerical constants. We favor the simplicity of the arguments over their precision regarding numerical constants. Here the parameter $s/65$ is not optimal (it can in fact be reduced to $s/2$). We give priority to the correct orders of magnitude and to the simplicity of the argument ahead of the exactness of the numerical constants.*

Proof. Let A be a matrix in $\mathbb{R}^{m\times N}$ satisfying $RIP(s)$. Let $v \in \mathrm{Ker}(A)\backslash\{0\}$. Let us show that $\|v\|_2 < \|v\|_1/(2\sqrt{s/65})$.

We decompose v according to the magnitude of the absolute values of its coordinates, into blocks of size s (cf. Figure 2). We recall that $v_1^* \geq v_2^* \geq \cdots \geq v_N^*$ is the nonincreasing rearrangement of $(|v_j|)_{j=1}^N$. We denote by I_1 the set of indices of the s largest coordinates of $(|v_j|)_{j=1}^N$, by I_2 the next s, and so on.



For every $j \geq 2$, one has

$$\left\|v_{I_j}\right\|_2 \leq \sqrt{s}\left\|v_{I_j}\right\|_\infty \leq \frac{\left\|v_{I_{j-1}}\right\|_1}{\sqrt{s}}. \tag{2.4}$$

We first treat the first block. By the condition $RIP(s)$, A acts as an isometry on all vectors of Σ_s; moreover $Av = 0$, and one then has

$$\|v_{I_1}\|_2 \leq 2\frac{\|Av_{I_1}\|_2}{\sqrt{m}} = \frac{2}{\sqrt{m}}\|A(v_{I_1} - v)\|_2 = \frac{2}{\sqrt{m}}\left\|A\Big(\sum_{j\geq 2} v_{I_j}\Big)\right\|_2 \leq 2\sum_{j\geq 2}\frac{\left\|Av_{I_j}\right\|_2}{\sqrt{m}} \leq 3\sum_{j\geq 2}\left\|v_{I_j}\right\|_2.$$

Finally, using relation (2.4) on each term of the previous sum, one obtains

$$\|v_{I_1}\|_2 \leq \frac{3}{\sqrt{s}} \sum_{j\geq 2} \left\|v_{I_{j-1}}\right\|_1 = \frac{3}{\sqrt{s}} \|v\|_1 .$$

We now treat the subsequent blocks. We apply formula (2.4) to each term:

$$\|v_{I_2} + \cdots\|_2 \leq \sum_{j\geq 2} \left\|v_{I_j}\right\|_2 \leq \frac{1}{\sqrt{s}} \sum_{j\geq 2} \left\|v_{I_{j-1}}\right\|_1 = \frac{\|v\|_1}{\sqrt{s}}.$$

One finally has:

$$\|v\|_2 \leq \|v_{I_1}\|_2 + \|v_{I_2} + \cdots\|_2 \leq \frac{4}{\sqrt{s}} \|v\|_1 < \frac{1}{2\sqrt{s/65}} \|v\|_1 .$$

■

Conclusion: We have proved the following chain of implications:

$$RIP(65s) \Longrightarrow Gelfand(s) \Longrightarrow NSP(s) \Longleftrightarrow RE(s).$$

It therefore suffices to construct a matrix satisfying $RIP(65s)$ in order to obtain a measurement matrix that allows the reconstruction of every vector of Σ_s by Basis Pursuit.

3 Sub-Gaussian random matrices and the RIP condition

In this section, we construct matrices satisfying $RIP(s)$ for an optimal number of measurements of order $m \sim s\log(eN/s)$. This number of measurements is optimal, since $RIP(s)$ implies $RE(c_0 s)$ (for an absolute constant c_0) and $RE(c_0 s)$ necessarily implies that $m \gtrsim s\log(eN/s)$. One therefore cannot construct a matrix satisfying $RIP(s)$ with fewer than $\sim s\log(eN/s)$ rows (up to an absolute constant).

It turns out that, to this day, the only matrices that have been constructed and that satisfy $RIP(s)$ for an optimal number of rows of order $s\log(eN/s)$ are random matrices.

The results we establish below are therefore of the type *with high probability, the matrix A satisfies $RIP(s)$*. They are thus results that hold only with high probability: one is not absolutely certain that the matrices constructed in this way satisfy RIP. Moreover, checking whether a given matrix satisfies RIP is an NP-hard problem.

It is somewhat disappointing to have results that hold only in probability; however:

1. the probabilities we shall obtain will be very close to 1
2. if one allows a number of measurements larger than $s\log(eN/s)$, then one can construct deterministic matrices satisfying $RIP(s)$ for $m = s^2$. If one wishes, and if it is possible, one may therefore accept a loss in the number of measurements to be taken (since $s^2 >> s\log(eN/s)$ in general) and consider only deterministic measurements.

Remark 3.1. *The construction of deterministic matrices satisfying $RIP(s)$ for a number of rows of order $s\log(eN/s)$ is one of the most difficult mathematical problems in CS. Such a construction would yield deterministic proofs of important results in mathematics (Dvoretzky's theorem, Kashin, etc.) which, for the moment, have only proofs that appeal to randomness.*

One must make precise what is meant by a deterministic matrix. First, a realization of a random matrix is not a deterministic construction. What one wants is a procedure that at no point appeals to randomness, that is able to fill in the $m \times N$ entries of the matrix A, and such that the output matrix satisfies $RIP(s)$ and $m \sim s \log(eN/s)$. This problem remains open to this day.

3.1 Gaussian random matrices and concentration

The standard example of random matrices is that of Gaussian matrices. They are built from a family of i.i.d. standard Gaussian random variables on $\mathbb{R}$.

Definition 3.2. *A standard Gaussian random matrix G is an $m \times N$ matrix*

$$G = \begin{pmatrix} g_{11} & \cdots & g_{1N} \\ \cdots & \cdots & \cdots \\ g_{m1} & \cdots & g_{mN} \end{pmatrix}$$

where $g_{11}, \ldots, g_{mN}$ are mN i.i.d. standard Gaussian random variables.

This definition is consistent with the classical one: a random vector G is Gaussian when its image under every linear form on this space is a Gaussian random variable on $\mathbb{R}$. Thus a **standard Gaussian random vector in** $\mathbb{R}^N$ is a vector in $\mathbb{R}^N$ whose coordinates are i.i.d. standard Gaussian random variables. The distribution of such a vector will be denoted $\mathcal{N}_N(0, I_N)$, where, here, the covariance matrix is I_N, the identity matrix of $\mathbb{R}^N$.

The only property of Gaussian random matrices that we shall use in this course is a deviation property.

To understand what a deviation property is, we first explain what a concentration property is.

A concentration result is stated as follows: for instance, if g is a real Gaussian random variable $\mathcal{N}(\mu, \sigma^2)$, then for every $t > 0$,

$$\mathbb{P}[|g - \mu| \geq t\sigma] \leq 2\exp(-t^2/2). \tag{3.1}$$

The idea is that if g is distributed according to the law $\mathcal{N}(\mu, \sigma^2)$, then with probability at least $1 - 2\exp(-t^2/2)$ it lies in the interval centered at μ and of length $2t\sigma$. The variable g is therefore "concentrated" around its mean μ.

Not every random variable is concentrated around its mean. Some do not even have a mean: the typical case is that of Cauchy random variables, which are absolutely continuous with respect to Lebesgue measure, with density $f(t) = [\pi(1 + t^2)]^{-1}$ for $t \in \mathbb{R}$, but which have no mean (one nevertheless says that they have a center of symmetry).

Other variables concentrate around their mean, but in some sense "less" than Gaussian variables. For example, a symmetric exponential random variable e, i.e., with density $f(t) = (2\sigma)^{-1}\exp(-|t - \mu|/\sigma)$ for $t \in \mathbb{R}$, satisfies: for every $t > 0$,

$$\mathbb{P}[|e - \mu| \geq t\sqrt{2}\sigma] = \exp(-\sqrt{2}t). \tag{3.2}$$

One says that e concentrates "as $\exp(-t)$" (around its mean, which equals μ here—its variance being $2\sigma^2$).

One can in fact find any type of concentration of the form "$\exp(-t^\alpha)$" for $0 < \alpha \leq 2$, or of the form "$1/t^p$" for $p \geq 1$. The behavior that is most sought after is the so-called **sub-Gaussian** behavior, i.e., of the form "$\exp(-t^2)$" as in (3.1). In this case one speaks of a ψ_2 variable, or

of a variable in L_{ψ_2}. One also says that a variable which concentrates as "$\exp(-t)$" is **sub-exponential**, because it has at least the same deviation behavior as an exponential variable. In this case, one speaks of a ψ_1 variable, or of a variable in L_{ψ_1}.

We now turn to the deviation property. Unlike concentration, deviation is concerned only with the behavior of the tail of the distribution, i.e., with $t \mapsto \mathbb{P}[|X| \geq t]$ when t is "large". In particular, the behavior of X around its mean is irrelevant to its deviation property. As for concentration, one speaks of **sub-Gaussian deviation** when, for every $t \geq c_0$, $\mathbb{P}[|X| \geq t] \leq 2\exp(-c_1 t^2)$, for $c_0 > 0$ large enough, and of **sub-exponential deviation** when $\mathbb{P}[|X| \geq t] \leq 2\exp(-c_1 t)$ for every $t \geq c_0$.

A rather useful way to characterize the type of deviation of a random variable is given by the growth rate of its moments.

Proposition 3.3. *Let X be a real random variable and let $\alpha \geq 1$. The following two properties are equivalent:*

1. *there exists an absolute constant $c_0 > 0$ such that for every $p \geq \alpha$, $\|X\|_{L_p} \leq c_0 p^{1/\alpha}$,*
2. *there exist two absolute constants c_1, c_2 such that for every $t \geq c_1$, $\mathbb{P}[|X| \geq t] \leq \exp(-t^\alpha / c_2^\alpha)$.*

Proof. Assume that the condition on the growth of the moments as $p^{1/\alpha}$ holds. By Markov's inequality, for every $t > 0$,

$$\mathbb{P}[|X| \geq t] \leq \inf_{p>0} \frac{\|X\|_{L_p}^p}{t^p} \leq \inf_{p \geq \alpha} \frac{c_0^p p^{p/\alpha}}{t^p} = \inf_{p \geq \alpha} \exp\Big(p \log\Big(\frac{c_0 p^{1/\alpha}}{t}\Big)\Big).$$

For $t \geq e c_0 \alpha^{1/\alpha}$, one obtains the desired result by taking $p = [t/(ec_0)]^\alpha$ in the previous inequality.

Conversely, assume that X has a deviation of the form "$\exp(-t^\alpha)$". We begin by showing that for $c = 2\max(c_1, c_2)$ one has $\mathbb{E}\exp[(|X|/c)^\alpha] \leq e$. We recall that for every $a > 0$,

$$\int_a^\infty \alpha t^{\alpha-1} \exp\left(-t^\alpha b\right) dt = \frac{1}{b} \exp\left(-b a^\alpha\right)$$

and that if Z is a nonnegative real random variable and $f : \mathbb{R}^+ \mapsto \mathbb{R}$ is a nondecreasing $\mathcal{C}^1$ function, then by Fubini

$$\mathbb{E}f(Z) - f(0) = \mathbb{E}\int_0^{+\infty} f'(t) I(t \leq Z) = \int_0^\infty f'(t)\mathbb{P}[Z \geq t] dt.$$

One then has:

$$\begin{aligned}
&\mathbb{E}\exp\Big(\frac{|X|^\alpha}{c^\alpha}\Big) - 1 = \int_0^{+\infty} \alpha u^{\alpha-1} \exp(u^\alpha)\mathbb{P}[|X| > cu] du \\
&\leq \int_0^{c_1/c} \alpha u^{\alpha-1}\exp(u^\alpha) du + \int_{c_1/c}^{+\infty} \alpha u^{\alpha-1} \exp\Big[u^\alpha\Big(1 - \frac{c^\alpha}{c_2^\alpha}\Big)\Big] du \\
&= \exp\Big(\frac{c_1^\alpha}{c^\alpha}\Big) - 1 + \frac{1}{(c/c_2)^\alpha - 1} \exp\Big[-\Big(\frac{c^\alpha}{c_2^\alpha} - 1\Big)\Big(\frac{c_1}{c}\Big)^\alpha\Big] \\
&\leq \exp\Big(\frac{c_1^\alpha}{c^\alpha}\Big) - 1 + \exp\Big[-\Big(\frac{c_1}{c}\Big)^\alpha\Big] \leq e - 1
\end{aligned}$$

since $\exp(x) + \exp(-x) \leq e$ when $0 \leq x \leq 1/2$. Moreover, for every integer q and every $x \geq 0$, $\exp(x) \geq x^q/q!$, hence

$$\mathbb{E}|X|^{\alpha q} = c^{\alpha q}\mathbb{E}\left[\left(\frac{|X|^\alpha}{c^\alpha}\right)^q\right] \leq c^{\alpha q} q! \mathbb{E}\exp\left(\frac{|X|^\alpha}{c^\alpha}\right) \leq eq!c^{\alpha q} \leq eq^q c^{\alpha q}.$$

Let $p \geq \alpha$. Let q be an integer such that $q\alpha \leq p < (q+1)\alpha$. One has:

$$\|X\|_{L_p} \leq \|X\|_{L_{(q+1)\alpha}} \leq e^{1/(q+1)\alpha} c(q+1)^{1/\alpha} \leq e^{1/p} c(2p/\alpha)^{1/\alpha} \leq 2ecp^{1/\alpha}.$$

■

A random variable therefore has sub-Gaussian (resp. sub-exponential) deviation if and only if its moments of order p grow no faster than $\sqrt{p}$ (resp. p).

In what follows, we shall be interested in squares of random variables, because the condition $RIP(s)$ involves $(1/m)\|Ax\|_2^2$, which is an average of m squares of random variables. We therefore need to know the deviation properties of variables of this type. To this end, we use the characterization of deviation by the growth rate of the moments given in Proposition 3.3.

Proposition 3.4. *Let X be a sub-Gaussian real random variable. Then X^2 is sub-exponential.*

Proof. By Proposition 3.3, since X is sub-Gaussian, for every $p \geq 2$ one has $\|X\|_{L_p} \leq c_0\sqrt{p}$, and therefore for every $q \geq 1$, $\left\|X^2\right\|_{L_q} = \|X\|_{L_{2q}}^2 \leq 2c_0^2 q$. Hence, by Proposition 3.3, X^2 is indeed sub-exponential. ■

There is a type of random variable that one expects to concentrate (and therefore to deviate) in a sub-Gaussian way. These are the empirical means of i.i.d. variables. Indeed, if $(X_m)_m$ is a sequence of i.i.d. variables possessing a moment of order 2, then, for $\bar{X}_m = \frac{1}{m}\sum_{i=1}^m X_i$, the CLT says that $\sqrt{m}(\bar{X}_m - \mathbb{E}X_1)$ is asymptotically Gaussian: for every $t > 0$, as $m \to +\infty$,

$$\mathbb{P}\left[\sqrt{m}|\bar{X}_m - \mathbb{E}X_1| \geq t\sigma\right] \longrightarrow \mathbb{P}[|g| > t]$$

where σ is the standard deviation of X_1. The deviations of $\sqrt{m}(\bar{X}_m - \mathbb{E}X_1)/\sigma$ are therefore similar to those of a standard Gaussian as $m \to +\infty$.

The CLT is an asymptotic result; it holds only for large values of m. The results we wish to establish in CS are for a fixed number of observations m. We wish to obtain concentration results for $\bar{X}_m$ at a fixed m. Such results are called non-asymptotic.

Proposition 3.5 (Bernstein's inequality for sub-exponential variables)**.** *Let $Z_1, \ldots, Z_m$ be i.i.d. real random variables with the same law as Z, such that for every $p \geq 1$,*

$$\|Z\|_{L_p} \leq C_0 p \tag{3.3}$$

where C_0 is an absolute constant.

Then there exists an absolute constant $c_1 > 0$ such that for every $t > 0$,

$$\mathbb{P}\left[\left|\frac{1}{m}\sum_{i=1}^m Z_i - \mathbb{E}Z\right| \geq tC_0\right] \leq 2\exp\left(-c_1 m \min(t^2, t)\right).$$

Proof. Without loss of generality, we assume that the Z_i are centered. Moreover,

$$\mathbb{P}[|\bar{Z}_m| \geq tC_0] = \mathbb{P}[\bar{Z}_m \geq tC_0] + \mathbb{P}[-\bar{Z}_m \geq tC_0]$$

so, replacing Z by $-Z$ if needed, it suffices to bound $\mathbb{P}[\bar{Z}_m \geq tC_0]$.

A classical way to prove concentration inequalities is to pass through the Laplace transform. By Markov's inequality,

$$\mathbb{P}\Big[\frac{1}{m}\sum_{i=1}^m Z_i \geq t\Big] \leq \inf_{\lambda>0} \frac{\mathbb{E}\exp(\lambda \bar{Z}_m)}{\exp(\lambda t)} = \inf_{\lambda>0}\Big(e^{-\lambda t}\prod_{i=1}^m \mathbb{E}\exp\Big(\frac{\lambda Z_i}{m}\Big)\Big). \tag{3.4}$$

We are thus led to bound $\mathbb{E}\exp(\mu Z)$ when Z is a centered sub-exponential variable and $\mu > 0$. Using the Taylor expansion of the exponential, one has

$$\mathbb{E}\exp(\mu Z) - 1 \leq \sum_{k\geq 2} \frac{\mu^k \mathbb{E}|Z|^k}{k!} \leq \sum_{k\geq 2} \frac{\mu^k C_0^k k^k}{k!}$$

and since $k! \geq \sqrt{2\pi k}(k/e)^k$, one has, for every $\mu > 0$ such that $e\mu C_0 < 1$,

$$\mathbb{E}\exp(\mu Z) \leq 1 + \sum_{k\geq 2}(e\mu C_0)^k = 1 + \frac{(e\mu C_0)^2}{1-(e\mu C_0)} \leq \exp\Big(\frac{(e\mu C_0)^2}{1-(e\mu C_0)}\Big)$$

since $1 + t \leq \exp(t)$ for every $t \in \mathbb{R}$.

One thus obtains, in (3.4),

$$\begin{aligned}\mathbb{P}\Big[\frac{1}{m}\sum_{i=1}^m Z_i \geq t\Big] &\leq \inf_{0<e\lambda C_0<m}\Big[e^{-\lambda t}\exp\Big(\frac{m[e(\lambda/m)C_0]^2}{1-[e(\lambda/m)C_0]}\Big)\Big] \\ &= \exp\Big[\inf_{0<e\lambda C_0<m}\Big(-\lambda t + \frac{(e\lambda C_0)^2}{m-(e\lambda C_0)}\Big)\Big] \leq \exp\Big[\inf_{0<e\lambda C_0<m/2}\Big(-\lambda t + \frac{2(e\lambda C_0)^2}{m}\Big)\Big]\end{aligned}$$

Now, if $t \leq 2eC_0$, we take $\lambda = mt/(2eC_0^2) \leq m/(2eC_0)$, and we then obtain

$$\mathbb{P}\Big[\frac{1}{m}\sum_{i=1}^m Z_i \geq t\Big] \leq \exp\Big(\frac{-mt^2}{4e^2C_0^2}\Big).$$

When $t > 2eC_0$, one has, for $\lambda = m/(2eC_0)$,

$$\mathbb{P}\Big[\frac{1}{m}\sum_{i=1}^m Z_i \geq t\Big] \leq \exp\Big(\frac{-mt}{4eC_0}\Big).$$

■

3.2 RIP is satisfied by Gaussian matrices

The previous section provides the probabilistic tools needed to prove that, with high probability, the condition $RIP(s)$ is satisfied by Gaussian random matrices for the minimal number of rows of order $m \sim s\log(eN/s)$.

Theorem 3.6. *Let $G \in \mathbb{R}^{m\times N}$ be a standard Gaussian random matrix. There exist two absolute constants c_0 and c_1 such that: with probability at least $1 - 2\exp(-c_0 m)$, for every $x \in \Sigma_s$,*

$$\frac{1}{2}\|x\|_2 \leq \frac{\|Gx\|_2}{\sqrt{m}} \leq \frac{3}{2}\|x\|_2$$

whenever $m \geq c_1 s\log(eN/s)$.

We shall in fact prove the result in a more general setting. We define

$$\Gamma = \frac{1}{\sqrt{m}} A \text{ where } A = \begin{pmatrix} X_1^\top \\ \vdots \\ X_m^\top \end{pmatrix}$$

and $X_1, \ldots, X_m$ are m i.i.d. random variables distributed as X, which is:

1. *isotropic*: $\mathbb{E}\langle X, x\rangle^2 = \|x\|_2^2$ for every $x \in \mathbb{R}^N$
2. *sub-Gaussian*: $\big\|\langle X, x\rangle\big\|_{L_p} \leq C_0 \|x\|_2 \sqrt{p}$ for every $x \in \mathbb{R}^N$ and every $p \geq 1$.

We show in what follows that Γ satisfies $RIP(s)$ with high probability; i.e., for this normalization, for every $x \in \Sigma_s$, $(1/2)\|x\|_2 \leq \|\Gamma x\|_2 \leq (3/2)\|x\|_2$.

The result does hold for Gaussian matrices, since if $\mathbf{g}$ is a standard Gaussian random vector in $\mathbb{R}^N$, then it is

1. isotropic: let $x \in \mathbb{R}^N$,

$$\mathbb{E}\langle \mathbf{g}, x\rangle^2 = \mathbb{E}\Big(\sum_{j=1}^N g_j x_j\Big)^2 = \sum_{j=1}^N x_j^2 = \|x\|_2^2$$

2. sub-Gaussian: for every $x \in \mathbb{R}^N$, $\langle \mathbf{g}, x\rangle$ has law $\mathcal{N}(0, \|x\|_2^2)$, hence $\big\|\langle \mathbf{g}, x\rangle\big\|_{L_p} = \|x\|_2 \|g\|_{L_p} \leq C_0 \|x\|_2 \sqrt{p}$ for every $p \geq 1$.

Before turning to the proof of RIP for sub-Gaussian matrices, we recall two results. The first is a volumetric argument concerning the complexity of convex sets.

Lemma 3.7. *Let s be an integer. Denote by S_2^{s-1} the Euclidean unit sphere of $\mathbb{R}^s$. Let $0 < \epsilon < 1$. There exists a set $\Lambda_\epsilon \subset S_2^{s-1}$ such that:*

1. *for every $x \in S_2^{s-1}$ there exists $y \in \Lambda_\epsilon$ such that $\|x - y\|_2 \leq \epsilon$,*
2. *$|\Lambda_\epsilon| \leq (5/\epsilon)^s$.*

Proof. Lemma 3.7 is proved by a volumetric argument. Take Λ_ϵ to be an ϵ-separated set (i.e., such that for every $x \neq y \in \Lambda_\epsilon$ one has $\|x - y\|_2 > \epsilon$) of maximal cardinality. The maximality of the cardinality of Λ_ϵ says that if one adds a point $z \in \mathcal{S}_2^{s-1} \backslash \Lambda_\epsilon$ to this set, i.e., if one considers $\Lambda_\epsilon \cup \{z\}$, then it no longer has the property of being ϵ-separated, which means that there necessarily exists an $x \in \Lambda_\epsilon$ such that $\|x - z\|_2 \leq \epsilon$. In particular, Λ_ϵ is an ϵ-net of $\mathcal{S}_2^{d-1}$ for ℓ_2^s. This proves the first property of Λ_ϵ. We now turn to its cardinality.

Since the points of Λ_ϵ are ϵ-separated, the balls $x + (\epsilon/2)B_2^s$ are disjoint and their union is contained in $(1 + \epsilon/2)B_2^s$. The sum of their volumes is therefore smaller than the volume of this last ball:

$$\text{vol}\left(\left(1 + \frac{\epsilon}{2}\right) B_2^s\right) \geq \sum_{x \in \Lambda_\epsilon} \text{vol}\left(x + \frac{\epsilon}{2} B_2^s\right).$$

By homogeneity of the volume, one obtains the result.

■

Lemma 3.8. *Let $B \in \mathbb{R}^{s\times s}$ be a symmetric matrix and let $0 < \epsilon < 1/2$. Let Λ_ϵ be an ϵ-net of S_2^{s-1} (the Euclidean unit sphere of $\mathbb{R}^s$). Then one has*

$$\|B\| = \sup_{x\in S_2^{s-1}} |\langle Bx, x\rangle| \leq (1-2\epsilon)^{-1} \max_{y\in\Lambda_\epsilon} |\langle By, y\rangle|$$

where we recall that $\|B\| = \sup_{\|x\|_2=1} \|Bx\|_2$ is the operator norm of $B : \ell_2^s \mapsto \ell_2^s$.

(We note that in the general case where B is not necessarily symmetric, the first inequality may fail to hold.)

Proof. Since B is symmetric, there exist an orthogonal matrix $P \in \mathcal{O}(s)$ and a diagonal matrix $D = \mathrm{diag}(\lambda_1, \ldots, \lambda_s)$ such that $B = PDP^\top$. One has

$$\begin{aligned}\|B\| &= \sup_{\|x\|_2=1} \|Bx\|_2 = \sup_{\|x\|_2=1} \|Dx\|_2 = \sup_{\|x\|_2=1} \sqrt{\sum_{j=1}^s \lambda_j^2 x_j^2} \\ &= \sup_{\|y\|_1=1} \sqrt{\sum_{j=1}^s \lambda_j^2 y_j} = \sqrt{\max\left(\lambda_1^2, \ldots, \lambda_s^2\right)} = \max\left(|\lambda_1|, \ldots, |\lambda_s|\right)\end{aligned}$$

and, on the other hand,

$$\begin{aligned}\sup_{x\in S_2^{s-1}} |\langle Bx, x\rangle| &= \sup_{\|x\|_2=1} |\langle DP^\top x, P^\top x\rangle| = \sup_{\|x\|_2=1} |\langle Dx, x\rangle| \\ &= \sup_{\|x\|_2=1} \left|\sum_{j=1}^s \lambda_j x_j^2\right| = \sup_{\|y\|_1=1} \left|\sum_{j=1}^s \lambda_j y_j\right| = \max\left(|\lambda_1|, \ldots, |\lambda_s|\right).\end{aligned}$$

One therefore has $\|B\| = \sup_{x\in S_2^{s-1}} |\langle Bx, x\rangle|$.

For the right-hand inequality, take $x \in S_2^{s-1}$ and $y \in \Lambda_\epsilon$ such that $\|x - y\|_2 \leq \epsilon$. One has

$$|\langle Bx, x\rangle| = |\langle By, y\rangle + \langle Bx, x-y\rangle + \langle B(x-y), y\rangle| \leq |\langle By, y\rangle| + 2\|x-y\|_2 \|B\|.$$

Since $\|x - y\|_2 \leq \epsilon$, the result follows. ∎

Proof of Theorem 3.6. We recall that Σ_s is the set of all s-sparse vectors of $\mathbb{R}^N$ and that S_2^{N-1} is the Euclidean unit sphere of ℓ_2^N. To show that A satisfies $RIP(s)$, it suffices to show that

$$\sup_{x\in\Sigma_s\cap S_2^{N-1}} \left|\, \|\Gamma x\|_2^2 - 1\right| \leq \frac{1}{4}. \tag{3.5}$$

One may write $\Sigma_s \cap S_2^{N-1} = \cup_{|J|=s} S^J$, where S^J is the set of all vectors of S_2^{N-1} whose support is contained in J.

By Lemma 3.7, for every $J \subset \{1, \ldots, N\}$ such that $|J| = s$, there exists Λ_ϵ^J, an ϵ-net of S^J of cardinality at most $(5/\epsilon)^s$. Now, applying Lemma 3.8 to the symmetric matrix $B = \Gamma_J^\top \Gamma_J - I_s$, where Γ_J is the submatrix of Γ whose column indices lie in J and I_s is the identity of $\mathbb{R}^s$, one has

$$\begin{aligned}&\sup_{x\in S^J} \left|\, \|\Gamma x\|_2^2 - 1\right| = \sup_{x\in S^J} \left|x_J^\top (\Gamma_J^\top \Gamma_J - I_s) x_J\right| = \sup_{x_J\in S_2^{s-1}} |\langle Bx_J, x_J\rangle| \\ &\leq (1-2\epsilon)^{-1} \max_{y\in\Lambda_\epsilon^J} |\langle By_J, y_J\rangle| = (1-2\epsilon)^{-1} \max_{x\in\Lambda_\epsilon^J} \left|\, \|\Gamma x\|_2^2 - 1\right|.\end{aligned}$$

Thus, one has

$$\sup_{x\in\Sigma_s\cap S_2^{N-1}} \big|\, \|\Gamma x\|_2^2 - 1\big| = \max_{|J|=s} \sup_{x\in S^J} \big|\, \|\Gamma x\|_2^2 - 1\big|$$
$$\leq \max_{|J|=s} \left[(1-2\epsilon)^{-1} \max_{x\in\Lambda_\epsilon^J} \big|\, \|\Gamma x\|_2^2 - 1\big|\right] \leq (1-2\epsilon)^{-1} \max_{x\in\Lambda_\epsilon} \big|\, \|\Gamma x\|_2^2 - 1\big| \tag{3.6}$$

where $\Lambda_\epsilon = \cup_{|J|=s}\Lambda_\epsilon^J$. In particular, the cardinality of Λ_ϵ satisfies

$$|\Lambda_\epsilon| \leq \binom{N}{s}\left(\frac{5}{\epsilon}\right)^s \leq \left(\frac{5eN}{s\epsilon}\right)^s \tag{3.7}$$

where we have used the inequality $\binom{N}{s} \leq (eN/s)^s$ (cf. exercise sheet).

We now apply Bernstein's inequality (cf. Proposition 3.5) at every point $x \in \Lambda_\epsilon$: for a given $x \in \Lambda_\epsilon$, with probability at least $1-2\exp(-c_0 m)$, one has $\big|\, \|\Gamma x\|_2^2 - 1\big| \leq 1/8$. Next, we use a union bound over Λ_ϵ: with probability at least $1-2|\Lambda_\epsilon|\exp(-c_0 m)$, for every $x \in \Lambda_\epsilon$, $\big|\, \|\Gamma x\|_2^2 - 1\big| \leq 1/8$. Then, when $(c_0/2)m \geq s\log(5eN/(s\epsilon))$, one has $\max_{x\in\Lambda_\epsilon} \big|\, \|\Gamma x\|_2^2 - 1\big| \leq 1/8$ with probability at least $1-2\exp(-(c_0/2)m)$. The result follows from (3.6) for $\epsilon = 1/4$. ∎

Conclusion: With high probability, standard Gaussian random matrices satisfy $RIP(s)$ for the optimal number of measurements (i.e., of rows) of order $m \sim s\log(eN/s)$. One may therefore use the rows of these matrices (i.e., i.i.d. standard Gaussian vectors in $\mathbb{R}^N$) as measurement vectors, and recover any s-sparse signal from its m projections onto these vectors by Basis Pursuit, i.e., by Linear Programming.

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